

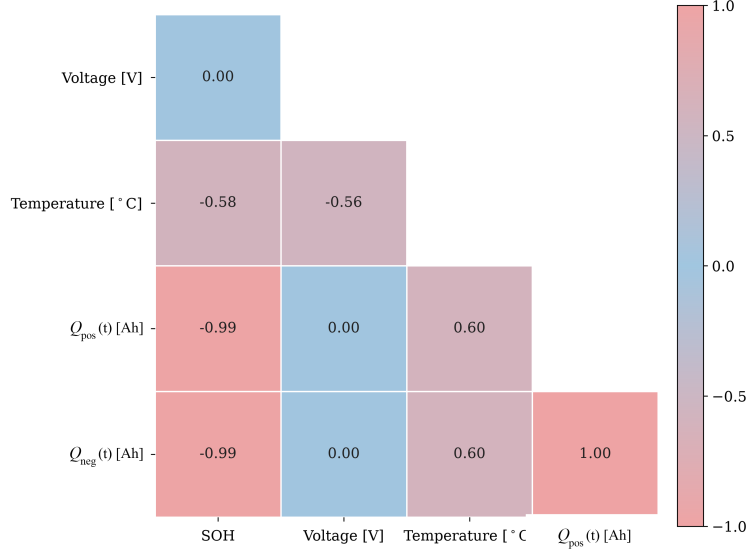


Figure 5.1: Pairwise Pearson correlation coefficients between SOH and the candidate input variables, including temperature and the separate cumulative currents $Q_{\text{pos}}(t)$ and $Q_{\text{neg}}(t)$. Only the non-redundant lower triangular part of the symmetric matrix is shown. The symmetric OKLAB scale maps zero correlation to blue and increasing absolute correlation magnitude toward red. Consequently, coefficients of -1 and $+1$ receive the same red endpoint colour, while the sign and exact value remain visible in the printed coefficient.

5.2.3 Formation of the Lag Sequence

Each learning sample is constructed as a lag sequence of voltage, temperature, and cumulative current, with the SOH at the current point in time used as the corresponding label, as illustrated in Figure 5.2. This transforms the continuous measurement stream into structured sliding-window samples that can be processed by a feedforward network without passing the complete life history of an individual cell directly into the model. Two parameters define the temporal representation. The time resolution Δt specifies the spacing between successive entries, while the number of lag steps L determines how many entries are retained. Their product, $L\Delta t$, gives the represented history duration.